COMPETITIVE ANALYSIS

COMPARATIVE ANALYSIS.

75% LESS CAPEX. 45% LESS OPEX.

A detailed comparison of Tobe Energy's electrolyzer against Alkaline, PEM, Solid Oxide, and Anion Exchange systems across cost, efficiency, and longevity.

>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

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EXECUTIVE SUMMARY

Tobe Energy's membrane-free hydrogen electrolyzer technology centers on **resonant electrolysis** — a proprietary approach eliminating ion-exchange membranes through frequency-optimized electrical waveforms. This architecture achieves **>92% system efficiency** at **$1.81/kg LCOH**, with deployment targeted for Q4 2026. The patent portfolio strategically secures core innovations while navigating existing IP barriers.

1. CORE INTELLECTUAL PROPERTY

1.1 Granted Patents (US)

PATENT NO.	TITLE	KEY CLAIMS	EXPIRY
US 11,228,765 B2	Resonant Electrolysis Cell Architecture	Membrane-free electrode configuration using pulsed DC at 45–60 Hz	2039
US 10,982,114 B2	Frequency-Optimized Gas Separation	Dynamic pressure modulation for H_2/O_2 separation without membranes	2040

1.2 Pending Applications

APPLICATION NO.	TECHNOLOGY AREA	STRATEGIC SIGNIFICANCE
US 17/842,301	Scalable Resonant Stack Design	Enables modular 1–10 MW systems; critical for cost reduction
US 18/015,442	Impurity-Tolerant Electrodes	Allows operation with non-purified water; reduces BoP costs by **18%**

2. COMPETITIVE IP LANDSCAPE

2.1 Key Players' Patent Clusters

COMPANY	FOCUS AREA	VULNERABILITY TO TOBE'S TECH
Nel Hydrogen	PEM membrane durability	High (membrane elimination bypasses core IP)
Siemens Energy	Alkaline system scaling	Medium (resonant approach achieves similar scale without membranes)
ITM Power	Grid integration	Low (complementary to Tobe's architecture)

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2.2 Freedom-to-Operate Analysis

- **Clear Path:** Zero overlapping claims in membrane-free electrolysis (validated by 2025 USPTO search)
- **Risk Zone:** Gas separation methods partially overlap with Siemens' EP 3,456,789; mitigated by Tobe's frequency modulation approach (**US 10,982,114 B2**)
- **White Space:** Resonant control algorithms represent untapped opportunity (>200 patent families analyzed)

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3. STRATEGIC IP ROADMAP

3.1 Near-Term (2024–2025)

- Finalize international PCT filings for **US 17/842,301** (priority deadline: Q3 2024)
- File continuation patents covering marine environment adaptations

3.2 Long-Term (2026+)

- Leverage resonant platform for ammonia synthesis (provisional filed Q1 2024)
- Build defensive portfolio around impurity tolerance (**US 18/015,442**) to block low-cost entrants

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CONCLUSION

Tobe Energy's patent portfolio establishes defensible dominance in membrane-free electrolysis — a **$4.2B whitespace market** per BloombergNEF 2025. The absence of membrane-related IP constraints accelerates

the path to **>92% efficiency** targets while creating licensing opportunities in industrial gas separation. Immediate priority: secure global coverage for resonant stack design before 2025 competitor filings.

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